Lab 2 - Mathematical Operations

Description

You must complete all the //TODO sections in template.cpp, which include implementing the following three mathematical functions:

1. Greatest Common Divisor.
2. Calculate the sum of all prime numbers less than a given number.
3. Convert a decimal number to its binary representation.
4. Sort the array data in ascending order.

Rules

- You must implement each of the three functions using for loop(s) or while loop(s) or both.
- You are strictly prohibited from using any built-in library functions (e.g., std::gcd from C++17) to perform these functions. Failure to comply will result in a score of 0.
- you can't modify anything outside of the //TODO sections in template.cpp.
- This assignment is 0-based.

Input Format

- The first line contains an integer n, representing the length of the vector<int> data.
- The second line contains n space-separated integers, representing the elements of the vector<int> data.
- The subsequent lines contain queries of the following types:
    - g a b: Calculate the Greatest Common Divisor of data[a] and data[b].
    - p a: Print the sum of all prime numbers less than data[a].
    - b a: Print the binary representation of data[a].
    - s: Sort the elements of data in ascending order.

Output Format

Print the result of each query on a new line.

Constraints

- 1 <= n <= 1000
- 0 <= a, b < n
- 0 <= data[i] <= 10^5

Sample Input

6

6 9 3 5 8 4

g 0 4

p 3

s

b 1

Sample Output

2

5

100